جامعة الإمام عبد الرحمن بن فيصل
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Project Phase 3 - Data Preparation

CIS 517: Data Mining and Data Warehousing

(First Semester)

Project Title: Predicting High-Risk Traffic Accident Locations Using Data Mining

Student Name	ID

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1. INTRODUCTION:

The preparation phase involves a systematic process of identifying and resolving inconsistencies, removing noise and irrelevant values, handling missing or outlier records, and reformatting variables to achieve a uniform structure. Emphasis is placed on maintaining the integrity of numeric, geographic, and environmental attributes, as these directly influence predictive accuracy in traffic-accident modeling.

To accomplish this, several cleaning and transformation actions were applied: outlier detection and removal to ensure statistical validity; correction of data-type inconsistencies across numeric and categorical fields; and precise formatting of numeric attributes particularly geographical coordinates, weather indicators, and distance measures to standardized decimal places. These steps enhance the dataset's reliability and facilitate efficient downstream tasks such as feature engineering and visualization in later phases.

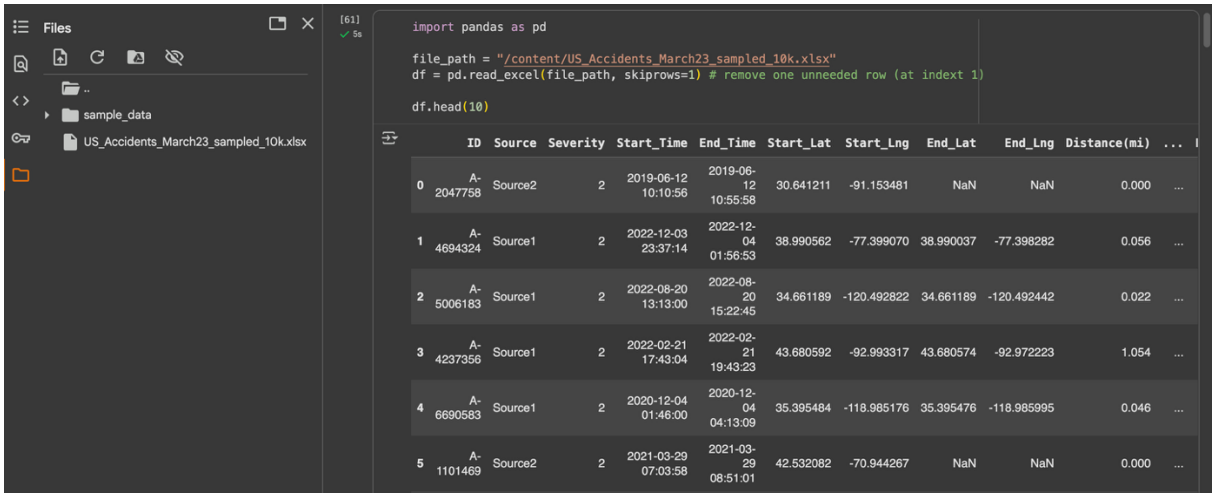
Overall, this phase establishes a robust foundation for subsequent data analysis and modeling, ensuring that all input variables are consistent, interpretable, and analytically sound.

2. METHODOLOGY: DATA CLEANING TECHNIQUES APPLIED

This section systematically presents the technical procedures employed during the data preprocessing phase. The primary objective was the transformation of the raw, heterogeneous accident dataset into a validated and normalized format to ensure optimal performance and robustness of the subsequent data mining models (Classification, Clustering, and Frequent Pattern Mining). The rationale for each operational step is carefully documented, laying a transparent and systematic foundation for subsequent analytical phases.

2.1 DATA LOADING AND INITIAL INSPECTION

The initial step involved loading the dataset into the computational environment using the pandas library, a fundamental tool in Python for data manipulation.
Code Implementation: The dataset was loaded using `pd.read_excel(file_path, skiprows=1)`. The `skiprows=1` argument was employed during ingestion to bypass extraneous header information and ensure accurate recognition of the actual column names.
Initial Verification: An initial inspection using df.head(10) was crucial for confirming successful data load and examining the general structure, column integrity, and initial data types. This inspection revealed an immediate data quality issue in the final column, Astronomical_Twilight, where extraneous semicolon delimiters (;;;...) were observed appended to the text values. This necessitated subsequent formatting efforts to address delimiter inconsistencies inherited from the raw data source.



	ID	Source	Severity	Start_Time	End_Time	Start_Lat	Start_Lng	End_Lat	End_Lng	Distance(mi)	...
0	A-2047758	Source2	2	2019-06-12 10:10:56	2019-06-12 10:55:58	30.641211	-91.153481	NaN	NaN	0.000	...
1	A-4694324	Source1	2	2022-12-03 23:37:14	2022-12-04 01:56:53	38.990562	-77.399070	38.990037	-77.398282	0.056	...
2	A-5006183	Source1	2	2022-08-20 13:13:00	2022-08-20 15:22:45	34.661189	-120.492822	34.661189	-120.492442	0.022	...
3	A-4237356	Source1	2	2022-02-21 17:43:04	2022-02-21 19:43:23	43.680592	-92.993317	43.680574	-92.972223	1.054	...
4	A-6690583	Source1	2	2020-12-04 01:46:00	2020-12-04 04:13:09	35.395484	-118.985176	35.395476	-118.985995	0.046	...
5	A-1101469	Source2	2	2021-03-29 07:03:58	2021-03-29 08:51:01	42.532082	-70.944267	NaN	NaN	0.000	...

Figure 1

2.2 HANDLING DUPLICATE RECORDS

Ensuring the uniqueness of observations is paramount in event-based data to prevent bias in frequency calculation and predictive model training.

Technique Rationale: The df.drop_duplicates() method was executed to identify and remove any rows that were exact duplicates across all 46 features. This operation ensures that each entry represents a singular, distinct recorded accident event.

Result: The procedural output confirmed that no duplicate rows were found in the sample dataset. This finding affirmed the high inherent uniqueness of the records, validating the dataset's fundamental integrity prior to downstream analysis.

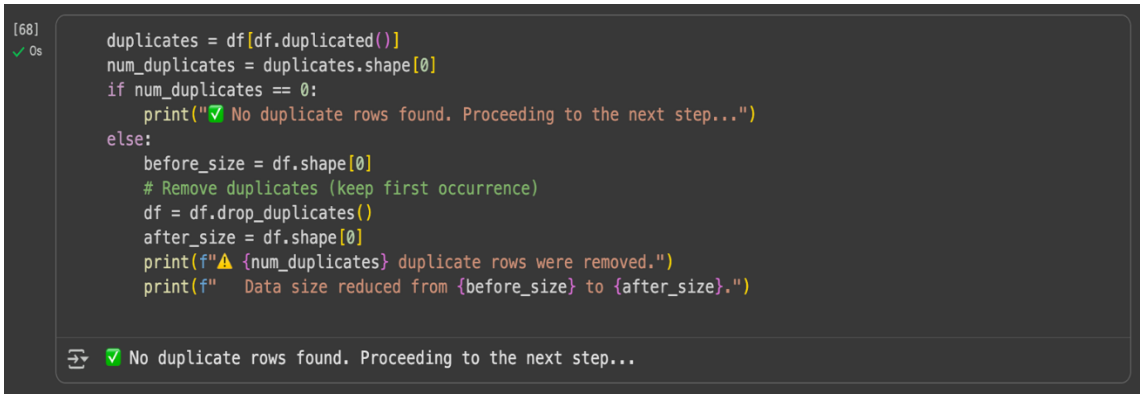


Figure 2

2.3 MISSING DATA IMPUTATION STRATEGY

The approach to handling missing values is designed based on data type to minimize statistical distortion across the feature space.

Numeric Columns Imputation (e.g., Geographic Coordinates, Weather Metrics): Missing values in numeric attributes (e.g., End_Lat, Temperature(F), Pressure(in)) were imputed using the column median. This strategy was employed because the median is a robust statistical measure that is less susceptible to displacement by extreme outlier values compared to the mean, thereby maintaining a more accurate representation of the typical central tendency.

Categorical Columns Imputation (e.g., Environmental Factors): For discrete or text-based features (e.g., Street, Weather_Condition, Sunrise_Sunset), missing entries were replaced with the mode (the most frequently occurring value). This method preserves the original distribution of categories, preventing misrepresentation in models that rely on class balance and frequency.

The following table summarizes key columns that underwent imputation:

Imputed Column	Imputation Statistic	Value Used
End_Lat	Median	36.150102
Wind_Chill(F)	Median	62.0
Pressure(in)	Median	29.85
Street	Mode	I-95 N
Weather_Condition	Mode	Fair
Sunrise_Sunset	Mode	Day

Table 1 key columns that underwent imputation

```
[63]
✓ 0s
    for col in df.columns:
        if df[col].isnull().any():  # Only process columns with missing data
            if pd.api.types.is_numeric_dtype(df[col]):
                median_value = df[col].median()
                df[col].fillna(median_value, inplace=True)
                print(f"   - Filled numeric column '{col}' with median value: {median_value}")
            else:
                mode_value = df[col].mode()[0]
                df[col].fillna(mode_value, inplace=True)
                print(f"   - Filled categorical column '{col}' with most frequent value: {mode_value}")

    print("\n✅ Missing data handling completed.")

    - Filled numeric column 'End_Lat' with median value: 36.150102000000004
    - Filled numeric column 'End_Lng' with median value: -87.9976635
    - Filled categorical column 'Street' with most frequent value: I-95 N
    - Filled categorical column 'Timezone' with most frequent value: US/Eastern
    - Filled categorical column 'Airport_Code' with most frequent value: KCQT
    - Filled categorical column 'Weather_Timestamp' with most frequent value: 2018-01-24 08:53:00
    - Filled numeric column 'Temperature(F)' with median value: 64.0
    - Filled numeric column 'Wind_Chill(F)' with median value: 62.0
    - Filled numeric column 'Humidity(%)' with median value: 68.0
    - Filled numeric column 'Pressure(in)' with median value: 29.85
    - Filled numeric column 'Visibility(mi)' with median value: 10.0
    - Filled categorical column 'Wind_Direction' with most frequent value: CALM
    - Filled numeric column 'Wind_Speed(mph)' with median value: 7.0
    - Filled numeric column 'Precipitation(in)' with median value: 0.0
    - Filled categorical column 'Weather_Condition' with most frequent value: Fair
    - Filled categorical column 'Sunrise_Sunset' with most frequent value: Day
    - Filled categorical column 'Civil_Twilight' with most frequent value: Day
    - Filled categorical column 'Nautical_Twilight' with most frequent value: Day

    ✅ Missing data handling completed.
```

Figure 3 MISSING DATA IMPUTATION STRATEGY OUTPUT

2.4 EXTREME OUTLIER MANAGEMENT

To enhance the stability and generalization performance of the derived predictive models, especially in location- and distance-based analysis, the dataset was purged of outliers.

- **Technique:** The Interquartile Range (IQR) method was applied to numeric columns to define boundary conditions. Critically, an extended multiplier of 5.0 × IQR was adopted (Bounds: Q1 − 5.0 × IQR and Q3 + 5.0 × IQR).
- **Methodological Justification:** This wider boundary was intentionally selected for a dataset involving road accidents. The use of a standard 1.5×IQR might exclude genuinely significant, albeit rare, high-severity events (e.g., accidents spanning long distances or occurring under highly unusual weather conditions). The 5.0×IQR threshold effectively retains meaningful extreme observations relevant to high-risk prediction while simultaneously eliminating data entry errors or statistically implausible values.

- **Impact:** This cleanup process resulted in a substantial reduction of the dataset, removing 2,790 rows. The data size was reduced from an initial 9,998 rows to a refined 7,208 rows, with columns like Distance(mi) and Visibility(mi)showing the most significant impact from the removal of extreme values.

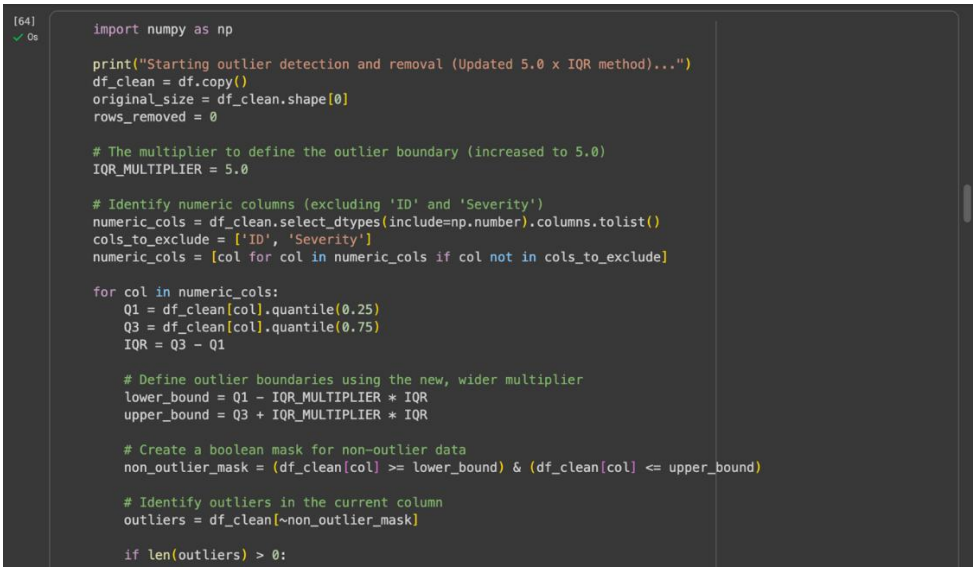


Figure 4 EXTREME OUTLIER MANAGEMENT OUTPUT

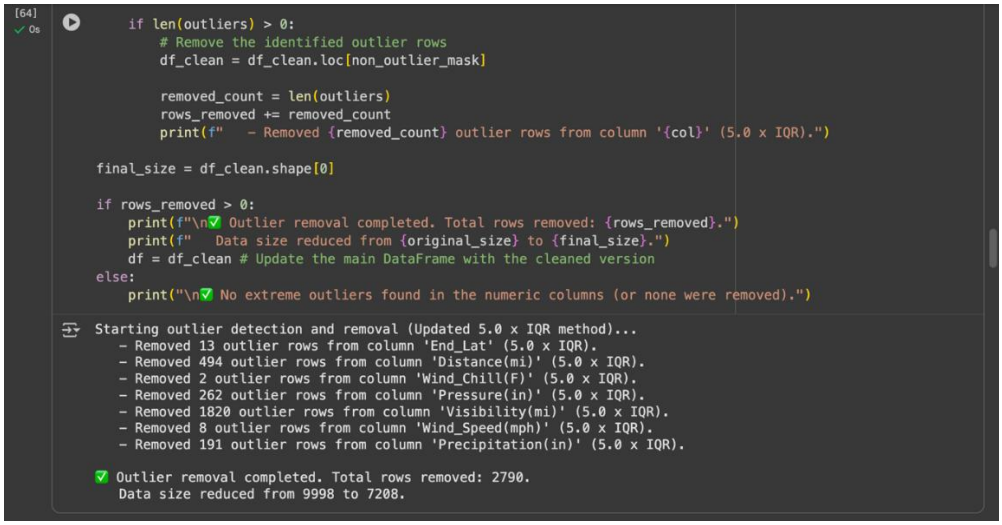


Figure 5

3. RESULTS AND DATA STANDARDIZATION

This final segment of the data preprocessing phase focuses on standardizing feature formats and summarizing the tangible outcomes of the cleaning operations, confirming the dataset's readiness for exploratory analysis and predictive modeling.

3.1 FEATURE STANDARDIZATION AND FORMATTING

Following imputation and outlier removal, a standardization step was performed to enforce consistent data types and decimal precision across all relevant numeric and geographic features. This consistency is essential for computational efficiency and uniformity in the downstream analytical models.

- **Objective:** The primary goal was to apply precise formatting to the cleaned floating-point numeric values. This action ensures data integrity is maintained, particularly for location coordinates, and guarantees consistency in data consumption by different tools or systems.
- **Decimal Precision Unification:** Specific formatting rules were applied to groups of features as detailed in the code:
    - **6 Decimal Places (.6f):** Applied to geographical coordinates (Start_Lat, Start_Lng, End_Lat, End_Lng). This high precision is necessary to accurately pinpoint accident locations.
    - **3 Decimal Places (.3f):** Applied specifically to the calculated distance field (Distance(mi)), maintaining sufficient resolution for travel distance measures.
    - **2 Decimal Places (.2f):** Applied to all weather and physical metric columns (Temperature(F), Wind_Chill(F), Humidity(%), Pressure(in), Visibility(mi), Wind_Speed (mph), Precipitation(in)). This standardizes environmental data while managing typical sensor precision levels.

3.2 FINAL DATA STATE SUMMARY

The culmination of the data cleaning workflow resulted in a significantly refined dataset. The primary impact was the reduction in the number of rows due to the removal of extreme outliers, while the dimensionality remained constant.

The table below provides a clear comparison of the dataset's state before and after the full cleaning and preprocessing sequence:

Metric	Initial State (After Load)	Final State (After Cleaning)	Change
Duplicate Records	0	0	Successfully confirmed unique records
Missing Values	18	0	Successfully imputed with median/mode
Extrem outliers values	2790	0	Successfully handeled outliers
Number of Rows	9,998	7,208	27.9% Reduction (2,790 rows removed)
Number of Columns	46	46	No Change

Table 2 comparison of the dataset's state before and after the full cleaning and preprocessing sequence

The final processed dataset, comprising 7,208 validated rows and 46 standardized columns, is now fit for the rigorous requirements of data mining tasks, having addressed issues related to missingness, extreme outliers, and format inconsistency.

3.3 SAVING THE CLEANED DATASET

The final step of the preprocessing phase was to persist with the clean data for subsequent use in modeling and feature engineering.

- **Action:** The resulting DataFrame was saved to a new Excel file.
- **Output Name:** To maintain clarity and version control, the file was named US_Accidents_March23_sampled_10k**_cleaned.xlsx**, clearly distinguishing the processed output from the raw input file. This ensures that the analytical process operates exclusively on the verified and high-quality data.

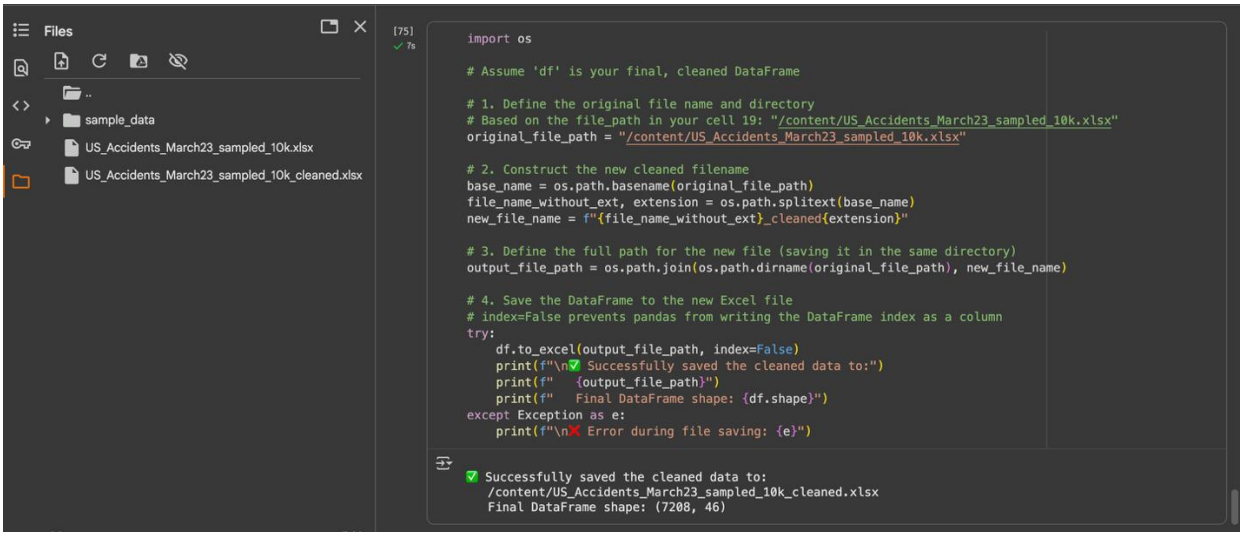

Files
..
sample_data
US_Accidents_March23_sampled_10k.xlsx
US_Accidents_March23_sampled_10k_cleaned.xlsx
[75]
7s
import os
Assume 'df' is your final, cleaned DataFrame
1. Define the original file name and directory
Based on the file_path in your cell 19: "/content/US_Accidents_March23_sampled_10k.xlsx"
original_file_path = "/content/US_Accidents_March23_sampled_10k.xlsx"
2. Construct the new cleaned filename
base_name = os.path.basename(original_file_path)
file_name_without_ext, extension = os.path.splitext(base_name)
new_file_name = f"{file_name_without_ext}_cleaned{extension}"
3. Define the full path for the new file (saving it in the same directory)
output_file_path = os.path.join(os.path.dirname(original_file_path), new_file_name)
4. Save the DataFrame to the new Excel file
index=False prevents pandas from writing the DataFrame index as a column
try:
df.to_excel(output_file_path, index=False)
print(f"\n✅ Successfully saved the cleaned data to:")
print(f" {output_file_path}")
print(f" Final DataFrame shape: {df.shape}")
except Exception as e:
print(f"\n❌ Error during file saving: {e}")
✅ Successfully saved the cleaned data to:
/content/US_Accidents_March23_sampled_10k_cleaned.xlsx
Final DataFrame shape: (7208, 46)

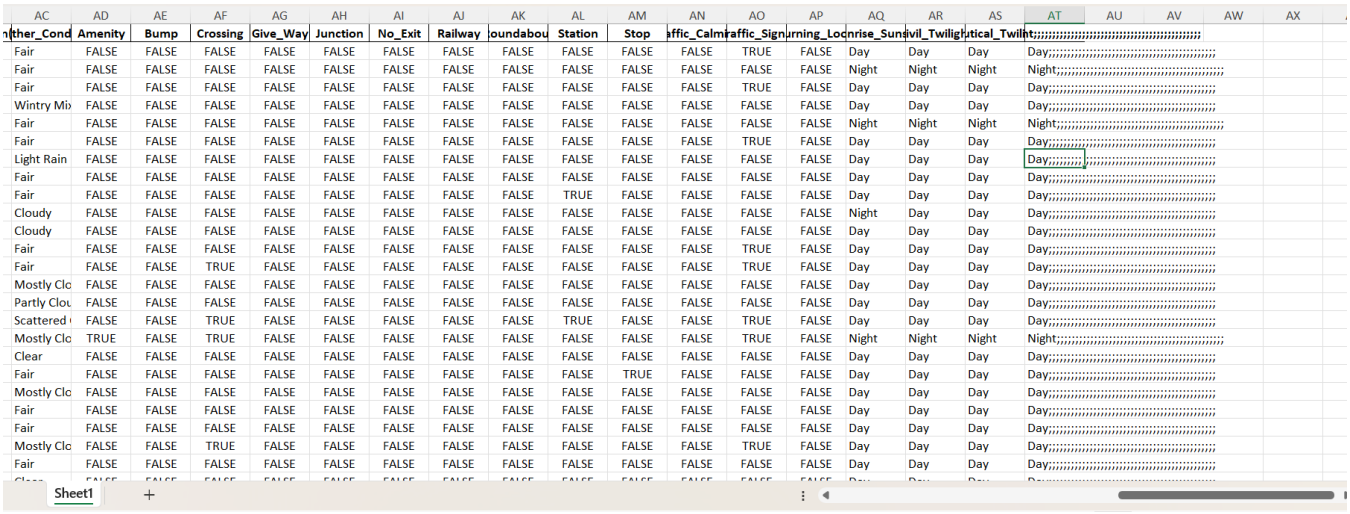
Figure 6

4. DATA TRANSFORMATION:

The transformations were designed to prepare the data for machine learning and analysis by cleaning, encoding, normalizing, and splitting the dataset while preserving the original structure and meaning of the data.

Step	Transformation Step	Description	Purpose	Affected Columns	Reason for Transformation	Result
1	Load Excel file	Read the raw accident data from Excel	Import data for processing	All columns	Initial data loading	DataFrame created from Excel
2	Remove trailing separators	Clean up corrupted string entries with excessive semicolons	Remove noise from string data	All string columns	Clean corrupted entries with excessive semicolons	Strings cleaned of trailing semicolon patterns
3	Drop 'Precipitation(in)' column	Remove irrelevant or empty column	Reduce dataset noise	Precipitation(in)	Column was irrelevant or empty	Column removed from dataset
4	Convert Boolean columns to numeric	Standardize Boolean values for modeling (TRUE → 1, FALSE → 0)	Prepare data for machine learning	Amenity, Bump, Crossing, Give_Way, Junction, No_Exit, Railway, Roundabout, Station, Stop, Traffic_Calming, Traffic_Signal, Turning_Loop	Standardize boolean values for modeling	TRUE/FALSE converted to 1/0
5	Convert datetime columns	Ensure proper datetime format for time-based analysis	Enable time-based filtering and analysis	Start_Time, End_Time, Weather_Timestamp	Ensure proper datetime format for time-based analysis	Converted to datetime objects
6	Ensure numeric columns	Coerce all numeric columns to proper types, handling errors	Prevent issues during modeling	All numeric columns	Prevent issues during modeling by enforcing numeric types	Numeric columns coerced to proper types

Table 3 Transformation Details Table



AC	AD	AE	AF	AG	AH	AI	AJ	AK	AL	AM	AN	AO	AP	AQ	AR	AS	AT
ther_Cond	Amenity	Bump	Crossing	Give_Way	Junction	No_Exit	Railway	oundabou	Station	Stop	affic_Calm	affic_Sign	rning_Loo	nrise_Suns	vil_Twilig	tical_Twili	ht;;;

AA	AB	AC	AD	AE	AF	AG	AH	AI	AJ	AK	AL	AM	AN	AO	AP	AQ	AR	AS	AT	AU
d_Speed(n	ther_Cond	Amenity	Bump	Crossing	Give_Way	Junction	No_Exit	Railway	oundabou	Station	Stop	affic_Calm	affic_Sign	urning_Loo	nrise_Suns	vil_Twilight	tical_Twil	nomical_Twilight		
11.5	Clear	0	0	0	0	0	0	0	0	0	0	0	0	0	Day	Day	Day	Day		
6	Cloudy	0	0	0	0	0	0	0	0	0	0	0	0	0	Day	Day	Day	Day		
8	Fair	0	0	0	0	0	0	0	0	0	0	0	0	0	Day	Day	Day	Day		
0	Fair	0	0	0	0	0	0	0	0	0	0	0	0	0	Night	Night	Night	Night		
8.1	Mostly Clo	0	0	0	0	0	0	0	0	0	0	0	0	0	Day	Day	Day	Day		
10.4	Mostly Clo	0	0	0	0	0	0	0	0	0	0	0	0	0	Night	Night	Day	Day		
11.5	Clear	0	0	0	0	0	0	0	0	0	0	0	0	0	Night	Night	Night	Night		
7	Cloudy	0	0	0	0	0	0	0	0	0	0	0	0	0	Day	Day	Day	Day		
9	Fair	0	0	0	0	0	0	0	0	0	0	0	0	0	Day	Day	Day	Day		
9.2	Overcast	0	0	0	0	0	0	0	0	0	0	0	0	0	Day	Day	Day	Day		
5	Fair	0	0	0	0	0	0	0	0	0	0	0	0	0	Night	Day	Day	Day		
3	Mostly Clo	0	0	0	0	0	0	0	0	0	0	0	0	0	Day	Day	Day	Day		
13	Fair	0	0	0	0	0	0	0	0	0	0	0	0	0	Day	Day	Day	Day		
9	Partly Clou	0	0	0	0	0	0	0	0	0	0	0	0	0	Day	Day	Day	Day		
0	Fair	0	0	0	0	0	0	0	0	0	0	0	0	0	Night	Night	Night	Night		
3.5	Clear	0	0	0	0	0	0	0	0	0	0	0	0	0	Day	Day	Day	Day		
9	Light Drizz	0	0	0	0	0	0	0	0	0	0	0	0	0	Night	Night	Night	Night		
7	Mostly Clo	0	0	0	0	0	0	0	0	0	0	0	0	0	Day	Day	Day	Day		
9	Fair	0	0	0	0	0	0	0	0	0	0	0	0	0	Day	Day	Day	Day		
0	Fair	0	0	0	0	0	0	0	0	0	0	0	0	0	Night	Night	Night	Day		
5	Mostly Clo	0	0	0	0	0	0	0	0	0	0	0	0	0	Day	Day	Day	Day		
12	Cloudy	0	0	0	0	0	0	0	0	0	0	0	0	0	Day	Day	Day	Day		
6	Fair	0	0	0	0	0	0	0	0	0	0	0	0	0	Night	Night	Night	Night		
8.1	Mostly Clo	0	0	0	0	0	0	0	0	0	0	0	0	0	Night	Night	Day	Day		

Figure 9 After transformations

A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P
ID	Source	Severity	Start_Time	End_Time	Start_Lat	Start_Lng	End_Lat	End_Lng	istance(m	Description	Street	City	County	State	Zipcode
-93381	Source2	2	2016-08-29 12:45:45	2016-08-29 13:27:00	33.88094	-117.603	36.1501	-87.9977	0	Left hand s	CA-91 W	Corona	Riverside	CA	92882
-899358	Source2	2	2021-09-15 08:42:49	2021-09-15 09:12:29	47.30233	-122.254	36.1501	-87.9977	0	Right lane	WA-18 W	Pacific	King	WA	98047
-579428(	Source1	2	2021-06-07 07:08:37	2021-06-07 09:05:38	45.1238	-93.3287	45.12573	-93.3462	0.859	Incident o	MN-610 W	Minneapol	Hennepin	MN	55443
-4657941	Source1	2	2022-06-27 23:15:13	2022-06-28 00:33:03	33.73122	-80.22	33.71882	-80.2303	1.041	Incident o	I-95 S	Alcolu	Clarendon	SC	29001
-2483841	Source2	2	2018-11-27 14:09:27	2018-11-27 14:39:04	33.74453	-117.981	36.1501	-87.9977	0	Lane block	Newland S	Midway Ci	Orange	CA	92655-133
-3099698	Source2	2	2017-12-01 17:18:51	2017-12-01 18:18:35	43.00554	-71.4767	36.1501	-87.9977	0	Accident o	Goffstown	Manchest	Hillsborou	NH	03102-273
-2217994	Source2	3	2019-03-13 04:47:21	2019-03-13 06:01:49	34.32065	-118.496	36.1501	-87.9977	0	Right hand	Foothill Fw	Sylmar	Los Angele	CA	91342
-7208223	Source1	3	2020-03-02 16:47:14	2020-03-02 17:16:27	32.81307	-96.9218	32.80463	-96.9218	0.583	At TX-356/	Loop 12 S	Irving	Dallas	TX	75060
-6775014	Source1	2	2020-12-01 13:52:00	2020-12-02 01:21:43	28.34391	-81.3627	28.34491	-81.364	0.107	Slow traffi	Buenaven	Kissimmee	Osceola	FL	34743
-215708	Source2	2	2016-09-01 07:48:43	2016-09-01 08:18:43	42.33034	-71.0614	36.1501	-87.9977	0	Accident o	I-93 S	Boston	Suffolk	MA	02118
-6573528	Source1	2	2021-02-03 17:45:35	2021-02-03 19:30:35	42.42273	-122.852	42.42321	-122.852	0.036	A crash ha	Crater Lak	White City	Jackson	OR	97503-162
-3712748	Source1	2	2022-12-01 15:33:10	2022-12-01 17:18:10	33.48078	-86.9209	33.46186	-86.9433	1.837	Incident o	I-59	Fairfield	Jefferson	AL	35064
-5257842	Source1	2	2022-09-27 08:30:16	2022-09-27 09:51:10	33.46319	-112.129	33.4641	-112.132	0.197	Incident o	N 33rd Av	Phoenix	Maricopa	AZ	85009-341
-925788	Source2	3	2021-08-25 17:54:40	2021-08-25 19:28:56	35.06686	-89.8441	36.1501	-87.9977	0	Right lane	Bill Morris	Memphis	Shelby	TN	38119
-6439921	Source1	2	2021-02-08 03:59:00	2021-02-08 05:39:09	33.66446	-112	33.6619	-112	0.177	Incident o	Piestewa F	Phoenix	Maricopa	AZ	85050
-3534877	Source1	3	2016-11-04 15:21:57	2016-11-04 21:21:57	36.15523	-115.161	36.1615	-115.16	0.437	At Charles	I-15 N	Las Vegas	Clark	NV	89102
-1479625	Source2	2	2020-05-23 23:13:34	2020-05-24 00:43:01	40.66596	-73.7569	36.1501	-87.9977	0	Left lane b	Belt Pkwy	Springfield	Queens	NY	11413
-6469036	Source1	2	2021-01-28 13:47:00	2021-01-28 14:58:00	34.07144	-118.125	34.07155	-118.117	0.452	Slow traffi	I-10 E	Alhambra	Los Angele	CA	91803
-1419375	Source2	2	2020-08-11 17:43:45	2020-08-11 18:29:04	30.52301	-90.9936	36.1501	-87.9977	0	Accident o	Magnolia I	Greenwell	East Bator	LA	70739-474
-4905570	Source1	2	2022-04-20 21:08:20	2022-04-20 22:26:29	35.01053	-81.8095	35.00985	-81.8102	0.061	Incident o	S Main St	Cowpens	Spartanbu	SC	29330-925
-5657415	Source1	2	2021-07-01 15:04:54	2021-07-01 17:15:45	33.97293	-80.3986	33.9734	-80.3992	0.049	Incident o	Somerset	Sumter	Sumter	SC	29150-182
-5955797	Source1	2	2021-10-30 13:00:22	2021-10-30 13:28:44	34.82062	-82.2804	34.82412	-82.2925	0.728	Stationary	Woodruff	Greenville	Greenville	SC	29607-574
-524354	Source2	3	2022-08-28 21:31:39	2022-08-28 22:15:38	41.04503	-73.5627	36.1501	-87.9977	1.5	Right lane	Governor J	Stamford	Western C	CT	06902
-2332330	Source2	2	2019-01-11 17:46:38	2019-01-11 18:16:13	47.46325	-122.27	36.1501	-87.9977	0	Right hand	WA-518 E	Seattle	King	WA	98188

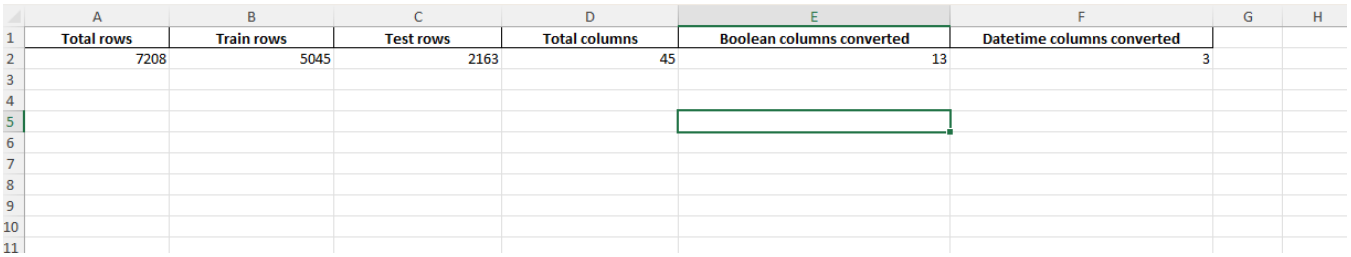
Figure 10 After transformations

5. DATA SPLITTING:

After transformation, the dataset was split into 70% training and 30% testing sets for model development.

STEP	DESCRIPTION	PURPOSE	CODE SNIPPET
7	Train/Test split	Split data into 70% training and 30% testing	train_df, test_df = train_test_split(df, test_size=0.3, random_state=42)
8	Save to Excel	Export transformed datasets for further use	train_df.to_excel(...), test_df.to_excel(...)
9	Create summary table	Summarize key metrics of transformation and split	summary_df = pd.DataFrame(summary)

Table 4 DATA SPLITTING



Total rows	Train rows	Test rows	Total columns	Boolean columns converted	Datetime columns converted
7208	5045	2163	45	13	3

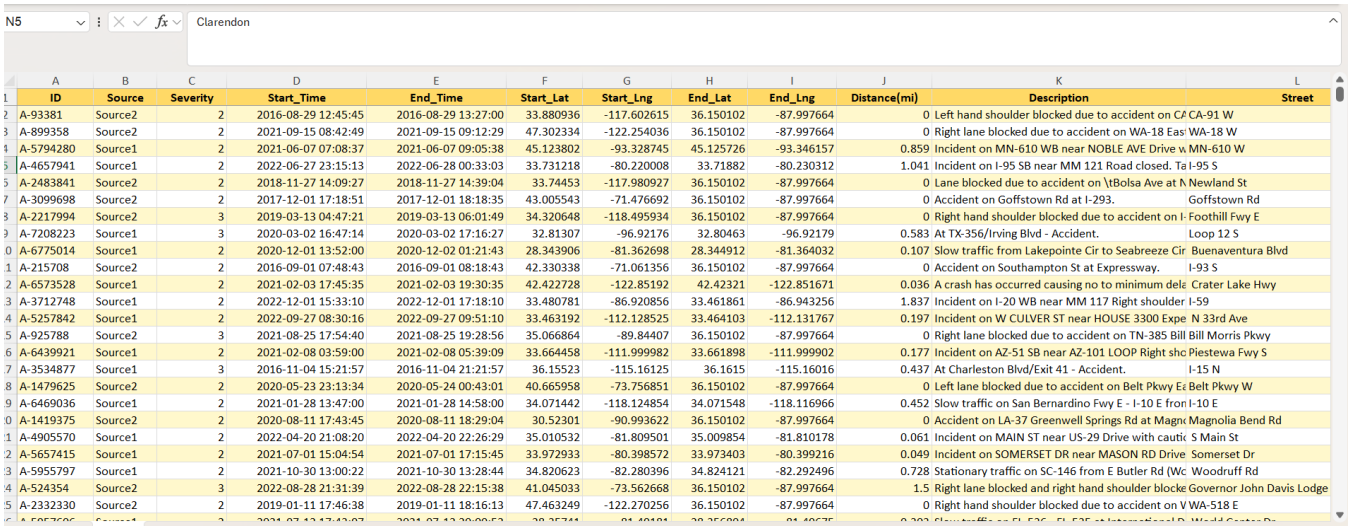
Figure 11 Summary Table from transformed_data_summary_V88.xlsx

Metric	Value
Total rows	7208
Train rows	5045
Test rows	2163
Total columns	45
Boolean columns converted	13
Datetime columns converted	3

Table 5 Summary Table from transformed_data_summary_V88.xlsx

6. EXCEL FORMATING:

This step exports the final cleaned and transformed dataset to a formatted Excel file.
The process includes saving the DataFrame into an Excel sheet and applying visual formatting using openpyxl, such as header highlighting, alternating row colours, light borders, and automatic column width adjustment.
The resulting Excel file provides clear, organised, and visually readable output that can be easily shared or used for further analysis.



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	A	B	C	D	E	F	G	H	I	J	K	L
1	ID	Source	Severity	Start_Time	End_Time	Start_Lat	Start_Lng	End_Lat	End_Lng	Distance(mi)	Description	Street
2	A-93381	Source2	2	2016-08-29 12:45:45	2016-08-29 13:27:00	33.880936	-117.602615	36.150102	-87.997664	0	Left hand shoulder blocked due to accident on CA	CA-91 W
3	A-899358	Source2	2	2021-09-15 08:42:49	2021-09-15 09:12:29	47.302334	-122.254036	36.150102	-87.997664	0	Right lane blocked due to accident on WA-18 Eas	WA-18 W
4	A-5794280	Source1	2	2021-06-07 07:08:37	2021-06-07 09:05:38	45.123802	-93.328745	45.125726	-93.346157	0.859	Incident on MN-610 WB near NOBLE AVE Drive w	MN-610 W
5	A-4657941	Source1	2	2022-06-27 23:15:13	2022-06-28 00:33:03	33.731218	-80.220008	33.71882	-80.230312	1.041	Incident on I-95 SB near MM 121 Road closed. Ta	I-95 S
6	A-2483841	Source2	2	2018-11-27 14:09:27	2018-11-27 14:39:04	33.74453	-117.980927	36.150102	-87.997664	0	Lane blocked due to accident on \tBolsa Ave at N	Newland St
7	A-3099698	Source2	2	2017-12-01 17:18:51	2017-12-01 18:18:35	43.005543	-71.476692	36.150102	-87.997664	0	Accident on Goffstown Rd at I-293.	Goffstown Rd
8	A-2217994	Source2	3	2019-03-13 04:47:21	2019-03-13 06:01:49	34.320648	-118.495934	36.150102	-87.997664	0	Right hand shoulder blocked due to accident on I-	Foothill Fwy E
9	A-7208223	Source1	3	2020-03-02 16:47:14	2020-03-02 17:16:27	32.81307	-96.92176	32.80463	-96.92179	0.583	At TX-356/Irving Blvd - Accident.	Loop 12 S
10	A-6775014	Source1	2	2020-12-01 13:52:00	2020-12-02 01:21:43	28.343906	-81.362698	28.344912	-81.364032	0.107	Slow traffic from Lakepointe Cir to Seabreeze Cir	Buenaventura Blvd
11	A-215708	Source2	2	2016-09-01 07:48:43	2016-09-01 08:18:43	42.330338	-71.061356	36.150102	-87.997664	0	Accident on Southampton St at Expressway.	I-93 S
12	A-6573528	Source1	2	2021-02-03 17:45:35	2021-02-03 19:30:35	42.422728	-122.85192	42.42321	-122.851671	0.036	A crash has occurred causing no to minimum dela	Crater Lake Hwy
13	A-3712748	Source1	2	2022-12-01 15:33:10	2022-12-01 17:18:10	33.480781	-86.920856	33.461861	-86.943256	1.837	Incident on I-20 WB near MM 117 Right shoulder	I-59
14	A-5257842	Source1	2	2022-09-27 08:30:16	2022-09-27 09:51:10	33.463192	-112.128525	33.464103	-112.131767	0.197	Incident on W CULVER ST near HOUSE 3300 Expe	N 33rd Ave
15	A-925788	Source2	3	2021-08-25 17:54:40	2021-08-25 19:28:56	35.066864	-89.84407	36.150102	-87.997664	0	Right lane blocked due to accident on TN-385 Bill	Bill Morris Pkwy
16	A-6439921	Source1	2	2021-02-08 03:59:00	2021-02-08 05:39:09	33.664458	-111.999982	33.661898	-111.999902	0.177	Incident on AZ-51 SB near AZ-101 LOOP Right sho	Piestewa Fwy S
17	A-3534877	Source1	3	2016-11-04 15:21:57	2016-11-04 21:21:57	36.15523	-115.16125	36.1615	-115.16016	0.437	At Charleston Blvd/Exit 41 - Accident.	I-15 N
18	A-1479625	Source2	2	2020-05-23 23:13:34	2020-05-24 00:43:01	40.665958	-73.756851	36.150102	-87.997664	0	Left lane blocked due to accident on Belt Pkwy Ea	Belt Pkwy W
19	A-6469036	Source1	2	2021-01-28 13:47:00	2021-01-28 14:58:00	34.071442	-118.124854	34.071548	-118.116966	0.452	Slow traffic on San Bernardino Fwy E - I-10 E fron	I-10 E
20	A-1419375	Source2	2	2020-08-11 17:43:45	2020-08-11 18:29:04	30.52301	-90.993622	36.150102	-87.997664	0	Accident on LA-37 Greenwell Springs Rd at Magno	Magnolia Bend Rd
21	A-4905570	Source1	2	2022-04-20 21:08:20	2022-04-20 22:26:29	35.010532	-81.809501	35.009854	-81.810178	0.061	Incident on MAIN ST near US-29 Drive with cautio	S Main St
22	A-5657415	Source1	2	2021-07-01 15:04:54	2021-07-01 17:15:45	33.972933	-80.398572	33.973403	-80.399216	0.049	Incident on SOMERSET DR near MASON RD Drive	Somerset Dr
23	A-5955797	Source1	2	2021-10-30 13:00:22	2021-10-30 13:28:44	34.820623	-82.280396	34.824121	-82.292496	0.728	Stationary traffic on SC-146 from E Butler Rd (Wo	Woodruff Rd
24	A-524354	Source2	3	2022-08-28 21:31:39	2022-08-28 22:15:38	41.045033	-73.562668	36.150102	-87.997664	1.5	Right lane blocked and right hand shoulder blocke	Governor John Davis Lodge
25	A-2332330	Source2	2	2019-01-11 17:46:38	2019-01-11 18:16:13	47.463249	-122.270256	36.150102	-87.997664	0	Right hand shoulder blocked due to accident on V	WA-518 E

Figure 12 EXCEL FORMATING

7. CHALLENGES AND PHASE SUMMARY

This section provides an overview of the main challenges faced during the data preparation phase and how each issue was handled to ensure the dataset was ready for further analysis.

6.1 CHALLENGES ENCOUNTERED AND MITIGATIONS

The data preparation stage was essential for building a reliable base for predictive modeling. However, several challenges were encountered while working with real-world accident data:

1. Extreme Outliers:

One major challenge was the large number of extreme outliers in some numerical features, especially Distance (mi) *and* Visibility (mi). Using the standard Interquartile Range (IQR) method would have removed many important records related to rare but severe accidents.
Mitigation: To address this, an extended 5.0 × IQR threshold was applied. This approach allowed us to keep meaningful extreme cases, such as long accident durations or unusual weather conditions, while still removing unrealistic data points. It helped maintain both accuracy and relevance in the dataset.

2. Missing Values in Key Features:

A noticeable amount of missing data appeared in important columns, such as geographic coordinates (End_Lat, End_Lng) and weather-related attributes.
Mitigation: Missing numerical values were replaced with the median to reduce the influence of outliers, while categorical features were filled using the mode. This approach completed the dataset without adding bias or altering its overall distribution.

3. Data Consistency and Formatting Issues:

Some columns showed inconsistent formats, including delimiter issues in the Astronomical_Twilight column and uneven decimal precision across numerical features.
Mitigation: Standard decimal formatting was applied 6 decimals for coordinates, 3 for distance, and 2 for environmental measures to keep data consistent and ready for modeling.

By the end of the data preprocessing phase, the raw dataset was successfully cleaned and transformed into a structured, high-quality analytical resource.
The main outcomes include:

- **Validated Dataset:** The final cleaned dataset contains 7,208 rows and 46 columns.
- **Data Integrity:** All duplicate records and missing values were completely removed.
- **Model Readiness:** Data transformation and splitting steps (such as feature engineering and train-test splitting) prepared the dataset for further analysis.

Overall, the dataset is now consistent, complete, and suitable for the next phase Exploratory Data Analysis (EDA) which will help identify key patterns and build predictive models for high-risk traffic accident areas.

REFERENCE:

[1] IBM Corporation, “Data Preparation in the Data Mining Process,” *IBM Documentation*, [Online]. Available: https://www.ibm.com/docs/en/db2/11.1.0?topic=studio-data-preparation-in-mining-process. [Accessed: Oct. 11, 2025].